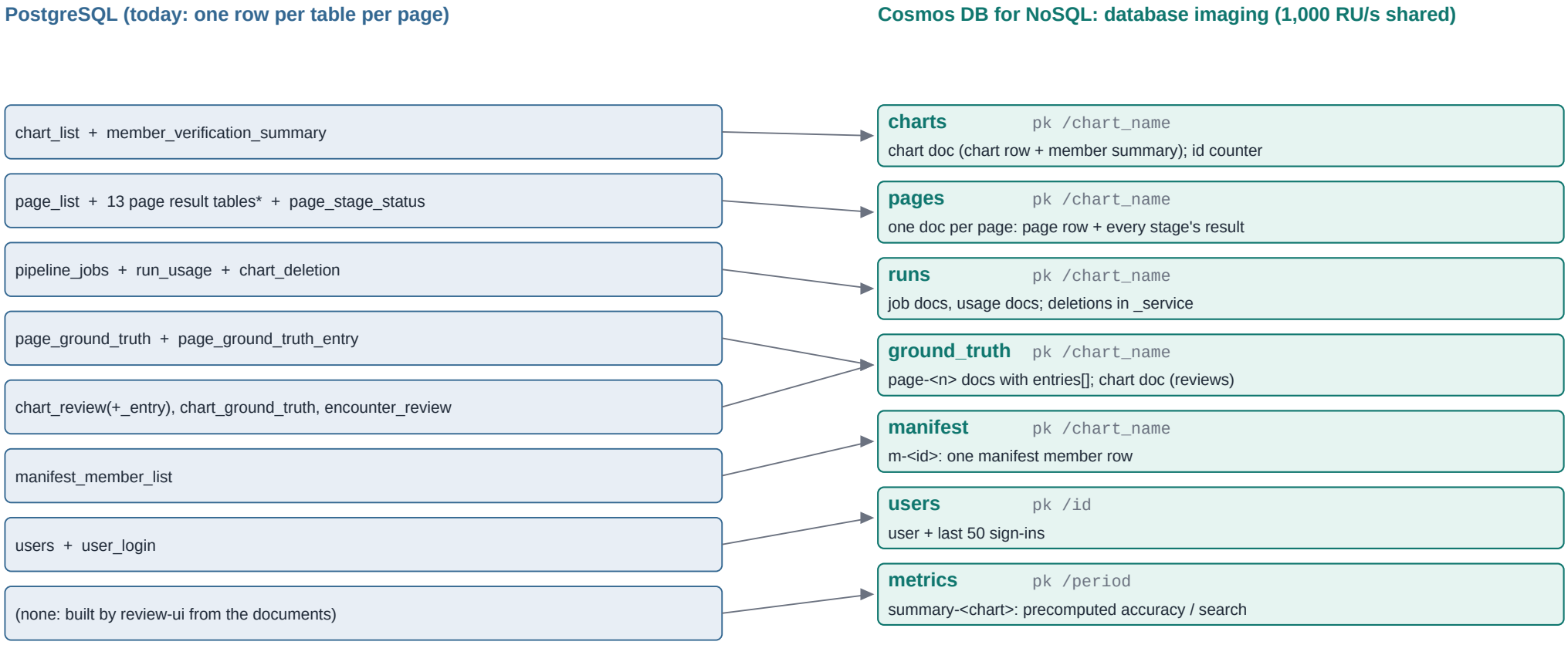
# Cosmos DB: containers, contents and the move from Postgres

Advantmed imaging pipeline · 14 Nov 2026 · database **imaging**, seven containers (infra/main.bicep) · written from the code: core-pipeline `common/db/cosmos_store.py`, review-ui `store/cosmos/`, `utilities/migrate_postgres_to_cosmos.py`

**In one line:** Postgres keeps one row per table per page and joins them; Cosmos keeps **one document per page** holding every stage's result for that page, grouped by chart (partition key `/chart_name`), so a chart is read in one partition with no joins. The row contents are the same column names and values; only the packaging changes.




* ocr_results, ocr_quality_results, blank_junk_classification, dos_extraction_results, page_classification, encounter_type_results, page_sequencing_results, page_continuity_results, additional_page_details, imaging_final, provider_signature_results, member_extraction_results (+ page_stage_status).

## 1. The seven containers and what each document holds

| Container | Partition key | Document id | What is inside | Written by |
|---|---|---|---|---|
| **charts** | /chart_name | chart<br>counters (pk _meta) | **chart**: the chart_list row (status, run / batch, tags, paths ...); **member_summary**: the member_verification_summary row. **counters**: next_id, so chart / page / job / manifest ids never collide. | core-pipeline |
| **pages** | /chart_name | page-<page_id> | **page** (page_list row) and one field per result table: **quality**, **ocr** {tesseract, docling, azuredocintel}, **blank_junk** {pass 1, pass 2}, **stages** {stage\|pass}, **dos**, **classification**, **encounter**, **sequencing**, **continuity**, **additional_details**, **imaging_final**, **provider_signature**, **member_extraction**. Indexing skips /ocr/*, /kv/*, /text/* (large text). | core-pipeline |
| **runs** | /chart_name<br>(_service when no chart) | job-<id><br>usage-<run><br>deletion-<id> | **job**: a pipeline_jobs row. **usage**: a run_usage row (time, pages, Doc Intel pages and cost, LLM tokens). **deletion**: a chart_deletion row (in the _service partition). | core-pipeline, review-ui |
| **ground_truth** | /chart_name | page-<n><br>chart | **page-<n>**: the page_ground_truth row (all label columns) with **entries[]** = its page_ground_truth_entry rows (entry_mode, labels, source_path, user_id, created_at). **chart**: chart_review (Reviewed mark), chart_ground_truth (member decision), encounter_review rows, review_entries (chart_review_entry). Keyed by chart name, so it survives a chart delete. | db-utilities, review-ui |
| **manifest** | /chart_name (= record_id) | m-<id> | **member**: one manifest_member_list row. | db-utilities, core-pipeline |
| **users** | /id | <users.id> | The users row (username, name, role, password hash, active, last login ...) and **logins**: the last 50 sign-ins (stands in for user_login). | review-ui |
| **metrics** | /period | summary-<chart><br>(period summary) | Precomputed per chart: the imaging document with its latest ground truth, the bulk-load ground truth and the Search-by-value rows (gzip). Rebuilt automatically when its fingerprint (the chart, ground-truth and manifest documents' etags) changes. Not migrated: review-ui builds it. | review-ui |

## 2. Where each Postgres table goes

| Postgres table (Part 1) | Cosmos container | Field in the document | Notes |
|---|---|---|---|
| chart_list | charts | chart | one chart doc per chart |
| member_verification_summary | charts | member_summary | |
| page_list | pages | page | one page doc per page |
| ocr_quality_results | pages | quality | incl. quality_detail.routing (Doc Intel routing row) |
| ocr_results | pages | ocr.{tesseract \| docling \| azuredocintel} | large text: not indexed |
| blank_junk_classification | pages | blank_junk.{1 \| 2} | pass 1 / pass 2 |
| page_stage_status | pages | stages.{stage\|pass} | |
| dos_extraction_results | pages | dos | dates[] kept as the JSONB array |
| page_classification | pages | classification | |
| encounter_type_results / page_sequencing_results / page_continuity_results | pages | encounter / sequencing / continuity | |

| Postgres table (Part 1) | Cosmos container | Field in the document | Notes |
|---|---|---|---|
| additional_page_details / provider_signature_results / member_extraction_results | pages | additional_details / provider_signature / member_extraction | |
| imaging_final | pages | imaging_final | Final values, confidences, review_fields |
| pipeline_jobs | runs | job (id job-<id>) | |
| run_usage | runs | usage (id usage-...) | the migration writes usage-pg-<id> |
| chart_deletion | runs (partition _service) | deletion | the migration writes deletion-pg-<id> |
| manifest_member_list | manifest | member (id m-<id>) | partition = record_id |
| page_ground_truth | ground_truth | page-<page_number> (the label columns) | |
| page_ground_truth_entry | ground_truth | page-<n>.entries[] | every write, oldest first |
| chart_review, chart_review_entry, chart_ground_truth, encounter_review | ground_truth | chart doc: chart_review, review_entries, chart_ground_truth, encounter_review | older databases may lack these tables: skipped |
| users, user_login | users | the user doc; logins[] (last 50) | |
| pipeline_stage | (not stored) | | a fixed registry: seeded in code |
| Part 2 tables (proposals) | (not stored) | | nothing writes them |

## 3. What changes, what stays the same

| | Postgres | Cosmos |
|---|---|---|
| Shape | one row per table per page; joins | one document per page; no joins (a chart is one partition) |
| Column names and values | snake_case columns | the same names and values inside the document fields |
| Ids | BIGSERIAL per table | the same ids, handed out from one counter document (taken in blocks with an etag check) |
| Writes | SQL statements | the whole changed document is upserted; an unchanged document costs nothing |
| Cross-chart lists (accuracy, search) | SQL over every chart | precomputed per-chart summaries in metrics |
| Ground truth history | page_ground_truth_entry rows | entries[] inside the page's ground-truth document |
| Switch | DB_BACKEND=postgres | DB_BACKEND=cosmos; both = write Postgres and copy every write to Cosmos |

## 4. How to migrate (upsert)

**Rule (14 Nov 2026):** the migration only **upserts**. For each chart it upserts every document (chart, pages, jobs, usage, manifest, ground truth, chart reviews), and only then removes that chart's documents Postgres no longer has (a page that is gone, a job or usage document with another id). The chart never disappears from Cosmos mid-copy, and running it again is always safe. Users are upserted, never removed. After each chart the counts are read back from Cosmos and compared with Postgres; a mismatch is reported and the run exits 1.

| Step | What | Command (Ubuntu VM, from core-pipeline with its venv) |
|---|---|---|
| 0 | Cosmos ready: database + seven containers (DBA, infra/main.bicep, or utilities/cosmos_containers.py create). Sign-in in core-pipeline/.env: COSMOS_ENDPOINT, COSMOS_AUTH (key / managed_identity / service_principal / entra) and its settings. | python ../utilities/cosmos_containers.py check |
| 1 | Dry run: Postgres counts per chart, nothing written. | python ../utilities/migrate_postgres_to_cosmos.py --all --dry-run |
| 2 | Full copy (upsert) of every chart, ground truth, reviews, usage, deletions, users. Sets the watermark for step 4. | python ../utilities/migrate_postgres_to_cosmos.py --all |
| 3 | Keep Cosmos current while both run: core-pipeline DB_BACKEND=both (each pipeline write is copied as it happens; a failed copy goes to logs/cosmos_mirror_failed.tsv). | DB_BACKEND=both in core-pipeline/.env |
| 4 | Incremental (e.g. nightly cron): charts changed since the last sync, review-ui / db-utilities / script edits, failed live copies. The watermark moves only when every count matches. | python ../utilities/migrate_postgres_to_cosmos.py --since-last |
| 5 | Point review-ui at Cosmos, compare a few charts and the Accuracy View with Postgres. | DB_BACKEND=cosmos in review-ui/.env |
| 6 | Cutover: core-pipeline DB_BACKEND=cosmos; Postgres read-only, then retired. | DB_BACKEND=cosmos in core-pipeline/.env |

Other options: `--chart <name>` (repeatable), `--changed-since 2026-11-10`, `--limit N`, `--skip-existing` (skip charts Cosmos already has), `--failed-file <path>`. Windows (Command Prompt): `cd core-pipeline && .venv\Scripts\activate.bat` then `python ..\utilities\migrate_postgres_to_cosmos.py --since-last`.

## 5. Incremental sync: what counts as changed

| Picked up when newer than the last sync | Copied |
|---|---|
| chart_list, page_list, any page result table, member_verification_summary, pipeline_jobs (updated_at) | the whole chart (upsert, then prune) |
| page_ground_truth, page_ground_truth_entry, manifest_member_list, chart_review(+_entry), chart_ground_truth, encounter_review, run_usage | that chart's ground truth, manifest, reviews and usage (a chart with ground truth only is copied too) |
| a line in logs/cosmos_mirror_failed.tsv (a DB_BACKEND=both copy that failed) | that chart; the line is cleared after the sync |
| users updated or a new sign-in | every user |
| chart_deletion rows | the deletion records |

The watermark is the Postgres time when the last successful sync started (`utilities/output/cosmos_sync_state.json`). A failed-list line that names no chart (?) is reported: run `--all` then.

## 6. Backfilling a ground-truth change (page_ground_truth)

To change a label in bulk, e.g. every blank / NA `is_invoice` to No: `utilities/gt_replace_value.py`. It changes **page_ground_truth** and adds a **page_ground_truth_entry** per changed page the way a bulk upload does (entry_mode bulk_upload; the page's latest bulk labels, else all its current labels, with the new value on top), in one transaction. Parameters at the top; dry run by default; prints counts and chart / page numbers only.

| Step | What | Command (Ubuntu VM) |
|---|---|---|
| 1 | Set COLUMN, FROM_VALUES ("BLANK" = NULL / empty; "NA", "N/A" ...), TO_VALUE, CHARTS ([] = all) at the top of the script. Run with DRY_RUN = True and check the counts. | python3 utilities/gt_replace_value.py |
| 2 | Set DRY_RUN = False and run again: Postgres page_ground_truth + page_ground_truth_entry change together. | python3 utilities/gt_replace_value.py |
| 3 | Copy the change to Cosmos: the page's updated_at and the new entry make it a changed chart; its ground_truth page documents are upserted with the new value and entries[]. | cd core-pipeline && python ../utilities/migrate_postgres_to_cosmos.py --since-last |
| 4 | review-ui picks it up: on Postgres at once; on Cosmos the chart's metrics summary is rebuilt on the next Accuracy View (its fingerprint changed). | |

Windows (Command Prompt): `python utilities\gt_replace_value.py`. The script works on Postgres; once Postgres is retired, the same change is made on the ground_truth documents.